Sir Syed CASE Institute of Technology

Lab #8 React

Web Engineering

SUBMITTED BY:

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ROLL NO:

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SUBMITTED TO:

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DATE:

16/04/2026

Lab Task

1. Install Node.js and verify the installation.
   Was already installed.

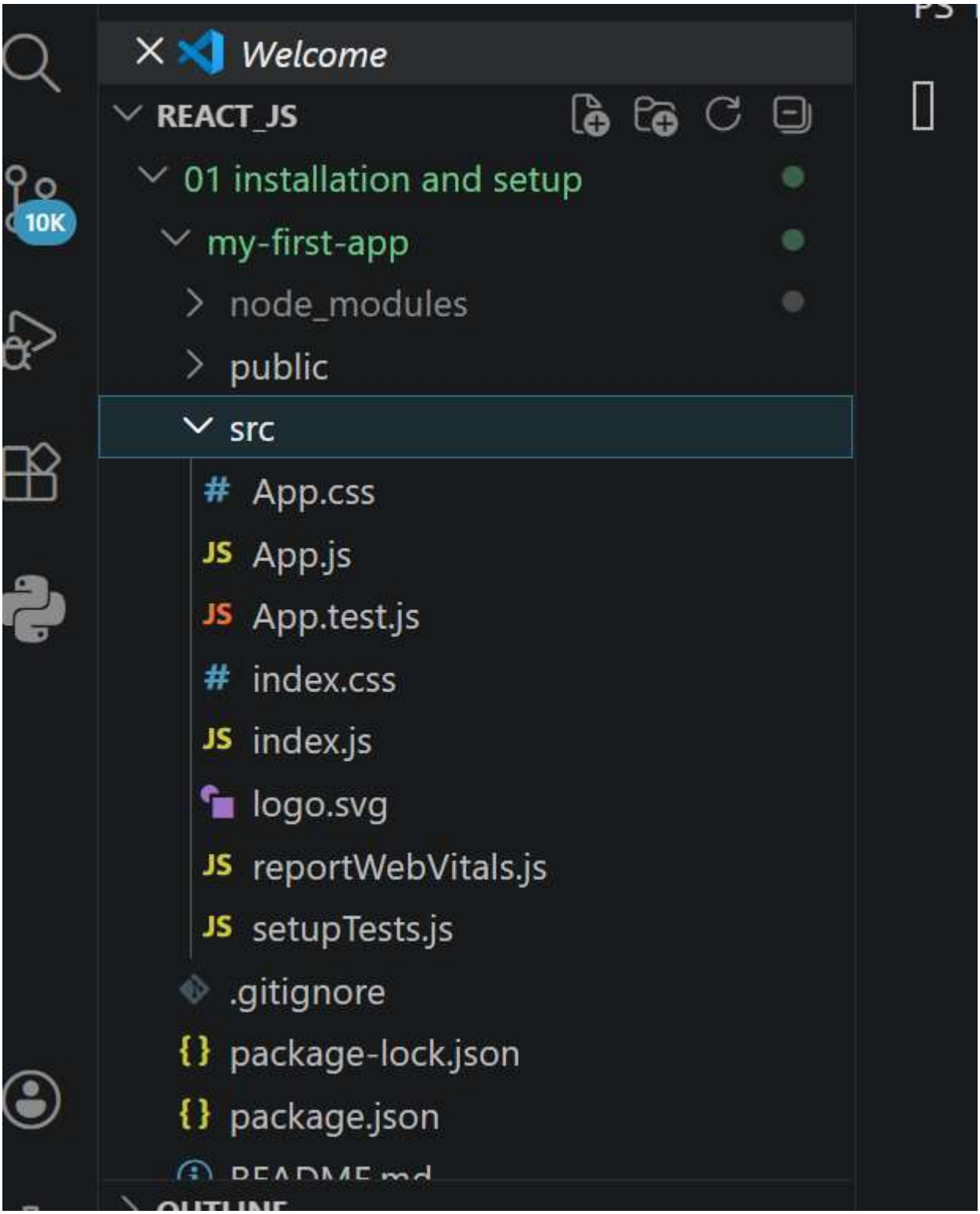
```
Command Prompt
Microsoft Windows [Version 10.0.19045.7058]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ahmad>node -v
v25.5.0

C:\Users\ahmad>_
```

2. Create a new React application.

```
PS D:\5th Semester data\Web Engineering\Lab 08 onward React\React_JS\01 installation and setup> npx create-react-app myFirstapp
PS D:\5th Semester data\Web Engineering\Lab 08 onward React\React_JS\01 installation and setup> npx create-react-app my-first-app
Need to install the following packages:
create-react-app@5.1.0
Ok to proceed? (y) y

Creating a new React app in D:\5th Semester data\Web Engineering\Lab 08 onward React\React_JS\01 installation and setup\my-first-app.

Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts with cra-template...


added 1298 packages in 7m

268 packages are looking for funding
  run `npm fund` for details

Installing template dependencies using npm...

added 17 packages, removed 1 package, and changed 1 package in 33s

267 packages are looking for funding
  run `npm fund` for details
Removing template package using npm...


removed 1 package, and audited 1314 packages in 21s
```

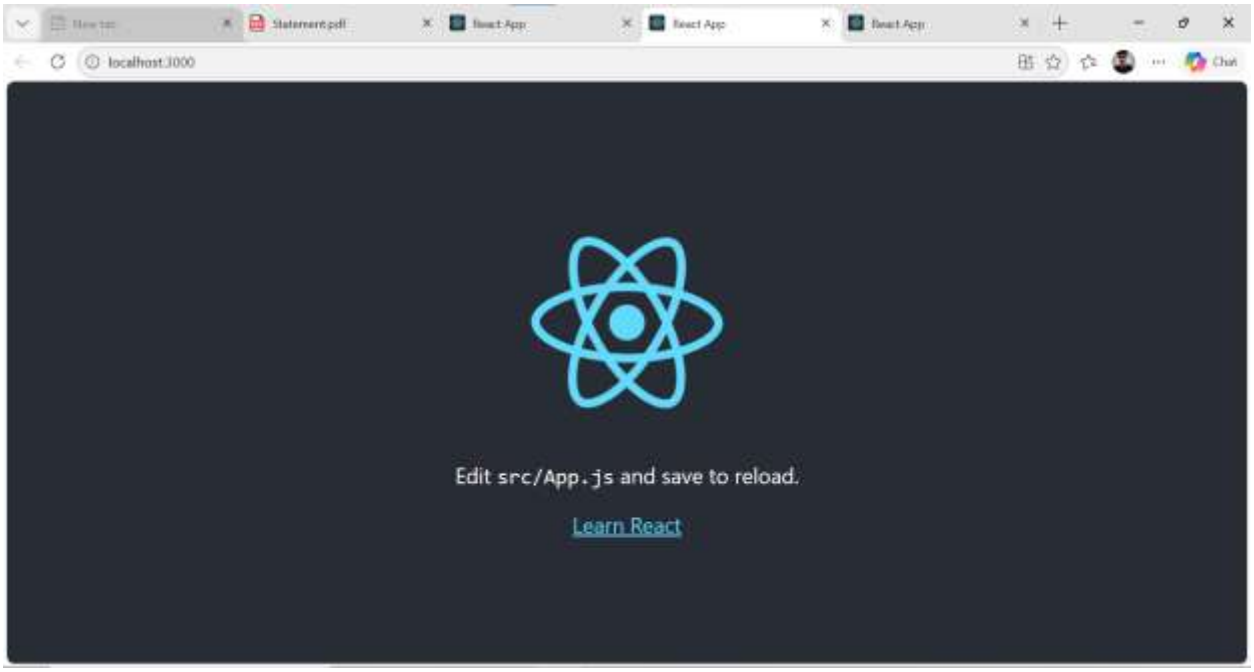
```
PS D:\5th Semester data\Web Engineering\Lab 08 onward React\React_JS\01 installation and setup> npx create-react-app my-first-app
267 packages are looking for funding
  run `npm fund` for details

26 vulnerabilities (9 low, 3 moderate, 14 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.

Success! Created my-first-app at D:\5th Semester data\Web Engineering\Lab 08 onward React\React_JS\01 installation and setup\my-first-ap
p
  npm start
    Starts the development server.

  npm run build
    Bundles the app into static files for production.

  npm test
    Starts the test runner.

  npm run eject
    Removes this tool and copies build dependencies, configuration files
```

3. Explore the folder structure of the React project.


Welcome
REACT_JS
01 installation and setup
my-first-app
node_modules
public
src
App.css
App.js
App.test.js
index.css
index.js
logo.svg
reportWebVitals.js
setupTests.js
.gitignore
package-lock.json
package.json

4. Run and test the application.

```
Success! Created my-first-app at D:\5th Semester data\Web Engineering\La
p
  npm start
    Starts the development server.

  npm run build
Compiled successfully!

You can now view my-first-app in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://172.25.224.1:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```



5. Install and configure Tailwind CSS

```
PS D:\5th Semester data\Web Engineering\Lab 08 onward React\React_JS\01 installation and setup\my-first-app> npm install -D tailwindcss@
3.4.0 postcss autoprefixer

added 3 packages, removed 2 packages, changed 1 package, and audited 1315 packages in 24s

268 packages are looking for funding
  run `npm fund` for details

26 vulnerabilities (9 low, 3 moderate, 14 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
PS D:\5th Semester data\Web Engineering\Lab 08 onward React\React_JS\01 installation and setup\my-first-app>
```

```
  npm audit fix --force

Run `npm audit` for details.
PS D:\5th Semester data\Web Engineering\Lab 08 onward React\React_JS\01 installation and setup\my-first-app> npx tailwindcss init -p
>>

Created Tailwind CSS config file: tailwind.config.js
Created PostCSS config file: postcss.config.js
PS D:\5th Semester data\Web Engineering\Lab 08 onward React\React_JS\01 installation and setup\my-first-app>
```

JS tailwind.config.js

01 installation and setup > my-first-app > JS tailwind.config.js

```js
/** @type {import('tailwindcss').Config} */
module.exports = {
  content: ["./src/**/*.{js,jsx,ts,tsx}"],
  theme: {
    extend: {},
  },
  plugins: [],
};

```

JS tailwind.config.js JS postcss.config.js

01 installation and setup > my-first-app > JS postcss.config.js

```js
module.exports = {
  plugins: {
    "@tailwindcss/postcss": {},
    autoprefixer: {},
  },
};

```

tailwind.config.js | JS postcss.config.js | # index.css 3

01 installation and setup > my-first-app > src > # index.css > body

```
/* stylelint-disable at-rule-no-unknown */
@tailwind base;
@tailwind components;
@tailwind utilities;
/* stylelint-enable at-rule-no-unknown */

body {
```

JS tailwind.config.js | JS postcss.config.js | # index.css 3 | JS App.js

01 installation and setup > my-first-app > src > JS App.js

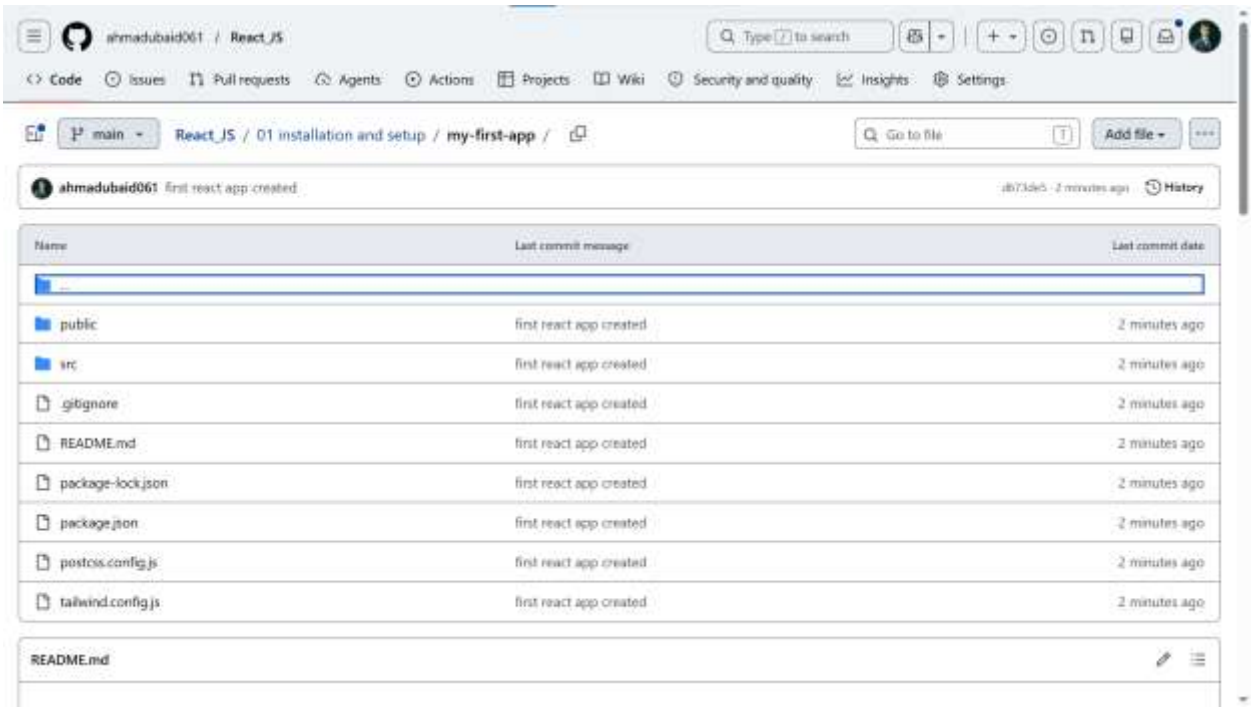
```
import logo from "./logo.svg";
import "./App.css";

function App() {
  return (
    <div className="App">
      <header className="App-header">
        <img src={logo} className="App-logo" alt="logo" />
        <h1 className="text-3xl font-bold text-blue-500">
          Hello, Tailwind Works
        </h1>
      </header>
    </div>
  );
}

export default App;
```

6. Configure and push files to your GitHub repository

```
MINGW64:/d/5th Semester data/Web Engineering/Lab 08 onward React/React_JS...

ahmad@DESKTOP-SOH5V33 MINGW64 /d/5th Semester data/Web Engineering/Lab 08 onward
 React/React_JS (main)
$ ls
'01 installation and setup'/   node_modules/       package.json
 README.md                     package-lock.json

ahmad@DESKTOP-SOH5V33 MINGW64 /d/5th Semester data/Web Engineering/Lab 08 onward
 React/React_JS (main)
$ cd '01 installation and setup'

ahmad@DESKTOP-SOH5V33 MINGW64 /d/5th Semester data/Web Engineering/Lab 08 onward
 React/React_JS/01 installation and setup (main)
$ git add .
warning: in the working copy of '01 installation and setup/my-first-app/.gitigno
re', LF will be replaced by CRLF the next time Git touches it
```

```
MINGW64:/d/5th Semester data/Web Engineering/Lab 08 onward React/React_JS...

ahmad@DESKTOP-SOH5V33 MINGW64 /d/5th Semester data/Web Engineering/Lab 08 onward
 React/React_JS/01 installation and setup (main)
$ git commit -m "first react app created"
[main d673de5] first react app created
 21 files changed, 17622 insertions(+)
 create mode 100644 01 installation and setup/Statement.pdf
 create mode 100644 01 installation and setup/my-first-app/.gitignore
 create mode 100644 01 installation and setup/my-first-app/README.md
 create mode 100644 01 installation and setup/my-first-app/package-lock.json
 create mode 100644 01 installation and setup/my-first-app/package.json
 create mode 100644 01 installation and setup/my-first-app/postcss.config.js
 create mode 100644 01 installation and setup/my-first-app/public/favicon.ico
 create mode 100644 01 installation and setup/my-first-app/public/index.html
```

```
ahmad@DESKTOP-SOH5V33 MINGW64 /d/5th Semester data/Web Engineering/Lab 08 onward
 React/React_JS/01 installation and setup (main)
$ git push origin main
Enumerating objects: 28, done.
Counting objects: 100% (28/28), done.
Delta compression using up to 4 threads
Compressing objects: 100% (27/27), done.
Writing objects: 100% (27/27), 770.55 KiB | 11.50 MiB/s, done.
Total 27 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ahmadubaid061/React_JS.git
   9bc960f..d673de5  main -> main

ahmad@DESKTOP-SOH5V33 MINGW64 /d/5th Semester data/Web Engineering/Lab 08 onward
 React/React_JS/01 installation and setup (main)
$
```



Repo Link:

https://github.com/ahmadubaid061/React_JS